5. 3 Diagonalization

- In many cases, the eigenvalue, eigenvector information contained with a matrix A can be displayed in a useful factorization of the form $A = PDP^{-1}$ where D is a diagonal matrix. The factorization enables us to compute A^k quickly for large values of k, a fundamental idea in several applications of linear algebra.

Ex: If $D = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}$ then $D^2 = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}\begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 5^2 & 0 \\ 0 & 3^2 \end{bmatrix}$

$D^3 = D\,D^2 = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}\begin{bmatrix} 5^2 & 0 \\ 0 & 3^2 \end{bmatrix} = \begin{bmatrix} 5^3 & 0 \\ 0 & 3^3 \end{bmatrix}$. In general, $D^k = \begin{bmatrix} 5^k & 0 \\ 0 & 3^k \end{bmatrix}$ for $k \geq 1$

- If $A = PDP^{-1}$ for some invertible P and diagonal D, then A^k is also easy to compute.

Ex: Let $A = \begin{bmatrix} 7 & 2 \\ -4 & 1 \end{bmatrix}$ Find a formula for A^k given that

$A = P\,D\,P^{-1}$ where $P = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix}$, $D = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}$

Sol $P^{-1} = \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}$. Then by associativity of matrix multiplication

$A^2 = (PDP^{-1})(PDP^{-1}) = PD(\underbrace{P^{-1}P}_{I})DP^{-1} = PDDP^{-1} = PD^2P^{-1} =$

$= \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix}\begin{bmatrix} 5^2 & 0 \\ 0 & 3^2 \end{bmatrix}\begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}$.

$A^3 = (PDP^{-1})A^2 = (PDP^{-1})PD^2P^{-1} = PD\underbrace{P^{-1}P}_{I}D^2P^{-1} = PDD^2P^{-1} = PD^3P^{-1}$

In general, for $k \geq 1$ $A^k = PD^kP^{-1} = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix}\begin{bmatrix} 5^k & 0 \\ 0 & 3^k \end{bmatrix}\begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}$

$= \begin{bmatrix} 2\cdot 5^k - 3^k & 5^k - 3^k \\ 2\cdot 3^k - 2\cdot 5^k & 2\cdot 3^k - 5^k \end{bmatrix}$

$AA = PBBP^{-1-1}$

- A square matrix A is said to be diagonalizable if A is similar to a diagonal matrix, that is, if for some invertible matrix P and diagonal matrix D.

Theorem:

The Diagonalization Theorem

An $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors.

In fact, $A = PDP^{-1}$, with D a diagonal matrix, if and only if the columns of P are n linearly independent eigenvectors of A. In this case, the diagonal entries of D are eigenvalues of A that correspond, respectively, to the eigenvectors in P.

- In other words, A is diagonalizable iff there are enough eigenvectors to form a basis of $\mathbb{R}^n$. We call such a basis an eigenvector basis of $\mathbb{R}^n$.

Ex: Diagonalize the following matrix, if possible

$A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}$. That is, find a invertible matrix P, and a diagonal matrix D such that

$A = PDP^{-1}$.

Given to us to avoid hard algebraic calc.

Sol Four steps:

1) Find the eigenvalues of A: $0 = \det(A - \lambda I) = -\lambda^3 - 3\lambda^2 + 4$

$= -(\lambda - 1)(\lambda + 2)^2$

$\Rightarrow \lambda = 1, \lambda = -2$

2) Find three linearly independent eigenvectors of A:

This is critical because if it fails, the above theorem says that A can not be diagonalized.

For $\lambda = 1 \Rightarrow (A - 1 \cdot I) \cdot x = 0 \Rightarrow \begin{bmatrix} 1-1 & 3 & 3 & 0 \\ -3 & -5-1 & -3 & 0 \\ 3 & 3 & 1-1 & 0 \end{bmatrix} \sim \begin{bmatrix} 0 & 3 & 3 & 0 \\ -3 & -6 & -3 & 0 \\ 3 & 3 & 0 & 0 \end{bmatrix}$

$$\sim \begin{bmatrix} 1 & 1 & 0 & 0 \\ -1 & -6 & -1 & 0 \\ 0 & 3 & 1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & -7 & -1 & 0 \\ 0 & 7 & 1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 7 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

x_3 free

$$\sim \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{aligned} x_1 + x_2 &= 0 \\ x_2 + x_3 &= 0 \\ x_3 &\text{ free} \end{aligned} \Rightarrow \begin{aligned} x_1 &= -x_2 \\ x_2 &= -x_3 \end{aligned}$$

$$x = \begin{bmatrix} x_3 \\ -x_3 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = v_1$$ ← Basis for $\lambda = 1$

Similarly basis for $\lambda = -2$ is $v_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$ $v_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$

Check and see that $\{v_1, v_2, v_3\}$ is linearly independent.

3) Construct P from the vectors in step 2. The order of vector is important.

$$P = \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix} = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

4) Construct D from the eigenvalues: The order of eigenvalues must match the order chosen for the columns of P.

$$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

verify $AP = PD$ i.e. $A = PDP^{-1}$ but $AP = PD$ is easier we don't need to calculate P^{-1})

Ex: Diagonalize $A = \begin{bmatrix} 2 & 4 & 3 \\ -4 & -6 & -3 \\ 3 & 3 & 1 \end{bmatrix}$ if possible.

Sol

Char. Eq. (given to us) $= -\lambda^3 - 3\lambda^2 + 4 = -(\lambda - 1)(\lambda + 2)^2$

Basis for $\lambda = 1$ $v_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

Basis for $\lambda = -2$ $v_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$

but no other eigenvalues and every eigenvector of A is a multiple of either v_1 or v_2.

Hence it is impossible to construct a basis for $\mathbb{R}^3$ using eigenvectors of A. By the above theorem, A is not diagonalizable.

Theorem: An $n \times n$ matrix with n distinct eigenvalues is diagonalizable.

- It is not necessary for an $n \times n$ matrix to have n distinct eigenvalues in order to be diagonalizable as one of the previous examples shows.

Ex: Determine if the following matrix is diagonalizable.

$$A = \begin{bmatrix} 5 & -8 & 1 \\ 0 & 0 & 7 \\ 0 & 0 & -2 \end{bmatrix}$$

Sol'n: The eigenvalues are 5, 0, -2 because the matrix is triangular. Since A is 3×3 matrix with three distinct eigenvalues, A is diagonalizable.

Matrices whose eigenvalues are not distinct

– If an $n \times n$ matrix A has n distinct eigenvalues with corresponding eigenvectors $v_1, v_2, \ldots, v_n$ and if $P = [v_1 \ \cdots \ v_n]$, then P is automatically invertible because its columns are linearly independent by the Theorem in page 5. When A is diagonalizable but has fewer than n distinct eigenvalues, it is still possible to build P in a way that makes P automatically invertible, as the next theorem shows.

Theorem:

Let A be an $n \times n$ matrix whose distinct eigenvalues are $\lambda_1, \ldots, \lambda_p$.

a. For $1 \leq k \leq p$, the dimension of the eigenspace for λ_k is less than or equal to the multiplicity of the eigenvalue λ_k.
b. The matrix A is diagonalizable if and only if the sum of the dimensions of the eigenspaces equals n, and this happens if and only if (*i*) the characteristic polynomial factors completely into linear factors and (*ii*) the dimension of the eigenspace for each λ_k equals the multiplicity of λ_k.
c. If A is diagonalizable and $\mathcal{B}_k$ is a basis for the eigenspace corresponding to λ_k for each k, then the total collection of vectors in the sets $\mathcal{B}_1, \ldots, \mathcal{B}_p$ forms an eigenvector basis for $\mathbb{R}^n$.

Ex: